

Linear Algebra

Inner Product Space

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Table of contents

01

Linear Form

02

Bilinear Form

03

Bilinear Form on Complex Vector Space

04

Inner Product

05

Inner Product Space

06

Norm Space

01

Linear Form

What are Linear Functions?

- $f: R^n \to R$ means that f is a function that maps real n-vectors to real numbers
- $f(x)$ is the value of function f at x (x is referred to as the argument of the function).
- $f(x) = f(x_1, x_2, \dots, x_n)$: where $x_1, x_2, \dots, x_n$ are arguments

Definition

A function $f: R^n \to R$ is linear if it satisfies the following two properties:

- Additivity: For any n-vector x and y, $f(x + y) = f(x) + f(y)$
- Homogeneity: For any n-vector x and any scalar $\alpha \in R$: $f(\alpha x) = \alpha f(x)$

Superposition property:

Definition

Superposition property:

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$$

Note

- A function that satisfies the superposition property is called linear

What are Linear Functions?

- If a function f is linear, superposition extends to linear combinations of any number of vectors:

$$f(\alpha_1 x_1 + \cdots + \alpha_k x_k) = \alpha_1 f(x_1) + \cdots + \alpha_k f(x_k)$$

Inner product is Linear Function?

Theorem 1

A function defined as the inner product of its argument with some fixed vector is linear.

Proof: $f(x) = a^T x = a_1 x_1 + a_2 x_2 + \cdots + a_n x_n$

1) $f(x+y) = a^\top (x+y) = a_1(x_1 + y_1) + \cdots + a_n(x_n + y_n)$

$= a_1 x_1 + \cdots + a_n x_n + a_1 y_1 + \cdots + a_n y_n = a^\top x + a^\top y = f(x) + f(y)$

2) $\alpha f(x) = \alpha(a^\top x) = \alpha(a_1 x_1 + \cdots + a_n x_n) = a_1 \alpha x_1 + \cdots + a_n \alpha x_n$

$= a^\top (\alpha x) = f(\alpha x)$

What are Linear Functions?

Theorem 2

If a function is linear, then it can be expressed as the inner product of its argument with some fixed vector.

Proof:

$$f(x) = f(x_1 e_1 + \cdots + x_n e_n) = x_1 f(e_1) + \cdots + x_n f(e_n)$$

$$= \begin{pmatrix} f(e_1) \\ \vdots \\ f(e_n) \end{pmatrix}^{\mathsf{T}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} f(e_1) \\ \vdots \\ f(e_n) \end{pmatrix}^{\mathsf{T}} x$$

What are Linear Functions?

Theorem 3

The representation of a linear function f in a specific basic as: $f(x) = a^T x$ is unique, which means that there is only one vector a for which $f(x) = a^T x$ holds for all x.

Proof:

$$f(x) = a^{\mathsf{T}} x\,, \qquad f(x) = b^{\mathsf{T}} x$$

$$f(e_1) = a^{\mathsf{T}} e_1 = a_1\,, \qquad f(x) = b^{\mathsf{T}} e_1 = b_1 \Rightarrow a_1 = b_1$$

$$f(e_2) = a^{\mathsf{T}} e_2 = a_2\,, \qquad f(x) = b^{\mathsf{T}} e_2 = b_2 \Rightarrow a_2 = b_2$$

$$\dots$$

$$f(e_n) = a^{\mathsf{T}} e_n = a_n\,, \qquad f(x) = b^{\mathsf{T}} e_n = b_n \Rightarrow a_n = b_n$$

$$\Rightarrow a = b$$

Linear Form Examples

Example

- Is average a linear function?
- Is maximum a linear function?

$$avg(x) = \frac{x_1 + \cdots + x_n}{n} = \frac{1}{n}x_1 + \cdots + \frac{1}{n}x_n = \begin{pmatrix} \frac{1}{n} \\ \vdots \\ \frac{1}{n} \end{pmatrix}^T x$$

$$x = \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} \max(x) = 5, \qquad y = \begin{pmatrix} 0 \\ 7 \\ 8 \end{pmatrix} \max(y) = 8$$

$$\max(x + y) = 11 \neq 5 + 8 = 13$$

02

Bilinear Form

Bilinear Form over a real vector space

Definition

Suppose V and W are vector spaces over the same field $\mathbb{F}$. Then a function $f: V \times W \rightarrow \mathbb{F}$ is called a **bilinear form** if it satisfies the following properties:

a) It is linear in its first argument:

i. $f(\mathbf{v_1} + \mathbf{v_2}, \mathbf{w}) = f(\mathbf{v_1}, \mathbf{w}) + f(\mathbf{v_2}, \mathbf{w})$ and

ii. $f(c\mathbf{v_1}, \mathbf{w}) = cf(\mathbf{v_1}, \mathbf{w})$ for all $c \in \mathbb{F}, \mathbf{v_1}, \mathbf{v_2} \in V$, and $\mathbf{w} \in W$.

b) It is linear in its second argument:

i. $f(\mathbf{v}, \mathbf{w_1} + \mathbf{w_2}) = f(\mathbf{v}, \mathbf{w_1}) + f(\mathbf{v}, \mathbf{w_2})$ and

ii. $f(\mathbf{v}, c\mathbf{w_1}) = cf(\mathbf{v}, \mathbf{w_1})$ for all $c \in \mathbb{F}, \mathbf{v} \in V$, and $\mathbf{w_1}, \mathbf{w_2} \in W$.

Bilinear Form

Note

Let V be a vector space over a field $\mathbb{F}$. Then the **dual** of V, denoted by V^*, is the vector space consisting of all linear forms on V.

Example

Let V be a vector space over a field $\mathbb{F}$. Show that the function $g: V^* \times V \rightarrow \mathbb{F}$ defined by

$$g(f, \mathrm{v}) = f(\mathrm{v}) \text{ for all } f \in V^*, \mathrm{v} \in V$$

is a bilinear form.

Example Solution!

- $g(f_1 + f_2, v) = (f_1 + f_2)v = f_1(v) + f_2(v) = g(f_1, v) + g(f_2, v)$
- $g(cf, v) = (cf)(v) = cf(v) = cg(f, v)$
- $g(f, v_1 + v_2) = f(v_1) + f(v_2) = g(f, v_1) + g(f, v_2)$
- $g(f, cv) = f(cv) = cf(v) = cg(f, v)$

Symmetric Bilinear Form

Definition

A **bilinear form** function $f: V \times V \rightarrow \mathbb{F}$ over a real vector space V is called symmetric if for all $v, w \in V$:

$$f(v, w) = f(w, v)$$

Bilinear Form arises from a matrix

Theorem 4

Every **bilinear form** function $f: V \times V \rightarrow \mathbb{F}$ over a real vector space V arises from a matrix for all $v, w \in V$:

$$f(v, w) = v^T A w$$

Proof?

$$f(v, w) = f\left(\sum_i v_i e_i, \sum_j w_j e_j\right) = \sum_i \sum_j v_i w_j f(e_i, e_j)$$

$$= \sum_i \sum_j v_i a_{ij} w_j = v^{\mathsf{T}} A w$$

Associated Matrices

Definition

If V is a finite-dimensional vector space, $B = \{b_1, \dots, b_n\}$ is a basis of V, and $f: V \times V \to \mathbb{F}$ be a **bilinear form** function the associated matrix A of f with respect to B is the matrix $[f]_B \in \mathbb{F}^{n \times n}$ whose (i,j)-entry is the value $f(b_i, b_j)$.

$$f(v,w) = v^T A w = v^T [f]_B w$$

$$[f]_{\mathcal{B}} = \begin{pmatrix} f(b_1,b_1) & \dots & f(b_1,b_n) \\ \vdots & & \vdots \\ f(b_n,b_1) & \dots & f(b_n,b_n) \end{pmatrix}$$

Associated Matrices

Note

The associated matrix changes if we use a different basis.

Example

For the bilinear form $f\left(\begin{bmatrix}a\\b\end{bmatrix},\begin{bmatrix}c\\d\end{bmatrix}\right) = 2ac + 4ad - bc$ on $\mathbb{F}^2$, find $[f]_B$ for basis $B = \{\begin{bmatrix}2\\1\end{bmatrix},\begin{bmatrix}-1\\4\end{bmatrix}\}$ and $[f]_P$ for basis P= $\{\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\}$

$$[f]_B = \begin{bmatrix}14 & 29\\-16 & 20\end{bmatrix} \qquad [f]_P = \begin{bmatrix}2 & 4\\-1 & 0\end{bmatrix}$$

03
Bilinear Form Over Complex Vector Space

Bilinear Form over a complex vector space

Definition

Suppose V and W are vector spaces over the same field $\mathbb{C}$. Then a function $f: V \times W \rightarrow \mathbb{C}$ is called a **bilinear form** if it satisfies the following properties:

a) It is linear in its first argument:

i. $f(\mathbf{v_1} + \mathbf{v_2}, \mathbf{w}) = f(\mathbf{v_1}, \mathbf{w}) + f(\mathbf{v_2}, \mathbf{w})$ and

ii. $f(\lambda \mathbf{v_1}, \mathbf{w}) = \lambda f(\mathbf{v_1}, \mathbf{w})$ for all $\lambda \in \mathbb{C}, \mathbf{v_1}, \mathbf{v_2} \in V$, and $\mathbf{w} \in W$.

b) It is conjugate linear in its second argument:

i. $f(\mathbf{v}, \mathbf{w_1} + \mathbf{w_2}) = f(\mathbf{v}, \mathbf{w_1}) + f(\mathbf{v}, \mathbf{w_2})$ and

ii. $f(\mathbf{v}, \lambda \mathbf{w_1}) = \bar{\lambda} f(\mathbf{v}, \mathbf{w_1})$ for all $\lambda \in \mathbb{C}, \mathbf{v} \in V$, and $\mathbf{w_1}, \mathbf{w_2} \in W$.

Bilinear Form over a complex vector space

Bilinear forms on $\mathbb{R}^n$	Bilinear forms on $\mathbb{C}^n$
Linear in the first variable	Linear in the first variable
Linear in the second variable	Conjugate linear in the second variable

04

Inner Product

Inner product over real vector space

Definition

An inner product is a positive-definite symmetric bilinear form.

An inner product on V is a function $\langle , \rangle : V \times V \rightarrow \mathbb{R}$ such that $v, w \in V, c \in \mathbb{R}$:

1. $\langle v, v \rangle = 0$ if and only if $v = 0$.
2. $\langle w, v \rangle = \langle v, w \rangle$.
3. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$.
4. $\langle cw, u \rangle = c\langle w, u \rangle$ for all $u, w \in V$ and $c \in \mathbb{R}$.
5. $\langle v, v \rangle \geq 0$ for all $v \in V$.

Inner Product

Why for bilinear form I wrote just two properties instead of four properties?

- Using properties (2) and (4) and again (2)

$$\langle w, cu\rangle = \langle cu, w\rangle = c\langle u, w\rangle = c\langle w, u\rangle$$

- Using properties (2), (3) and again (2)

$$\langle w, u+v\rangle = \langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle = \langle w, u\rangle + \langle w, v\rangle$$

1. $\langle v, v\rangle = 0$ if and only if $v = 0$.
2. $\langle w, v\rangle = \langle v, w\rangle$.
3. $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$ for all $u, v, w \in V$.
4. $\langle cw, u\rangle = c\langle w, u\rangle$ for all $u, w \in V$ and $c \in \mathbb{R}$.
5. $\langle v, v\rangle \geq 0$ for all $v \in V$.

Inner Products

Note

- For $v \in V, \langle 0, v \rangle = 0$, $\langle v, 0 \rangle = 0$.

$$\langle 0, v \rangle = \langle 0u, v \rangle = 0\langle u, v \rangle = 0$$

General Inner product

Definition

Suppose that $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ and that V is a vector space over $\mathbb{F}$. Then an **inner product** on V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ such that the following three properties hold for all $c \in \mathbb{F}$ and all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in V$:

a) $\langle \mathbf{v}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{v} \rangle}$ (conjugate symmetry)

b) $\langle \mathbf{v}+cx, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle + c\langle x, \mathbf{w} \rangle$ (linearity)

c) $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$. (pos. definiteness)

Inner Products for vectors

Note

- The standard inner product between vectors is: ($x, y \in \mathbb{R}^n$)

$$\langle x, y \rangle = x^T y = \sum x_i y_i$$

- The function $\langle \cdot, \cdot \rangle : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ defined by

$$\langle \mathrm{v}, \mathrm{w} \rangle = v^* \mathrm{w} = \sum_{i=1}^{n} \bar{v}_i \, w_i$$

for all $\mathrm{v}, \mathrm{w} \in \mathbb{C}^n$ is an inner product on $\mathbb{C}^n$.

Inner Products for matrices

Note

The standard inner product between two matrices is: $(X, Y \in \mathbb{R}^{m \times n})$

$$\langle X, Y \rangle = trace(X^T Y) = \sum_i \sum_j X_{ij} Y_{ij}$$

Example

Find the inner products of following matrices:

$$U = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \qquad V = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\langle U, V \rangle = trace(U^\top V) = trace\left(\begin{bmatrix} 2 & 0 \\ 2 & -2 \end{bmatrix}\right) = 0$$

Inner Product for functions

Note

Let $a < b$ be real numbers and let $C[a,b]$ be the vector space of continuous functions on the real interval $[a,b]$. The function $\langle \cdot,\cdot \rangle : C[a,b] \times C[a,b] \to \mathbb{R}$ defined by

$$\langle f,g \rangle = \int_a^b f(x)g(x)\,dx \qquad \text{for all} \qquad f,g \in C[a,b]$$

is and inner product on $C[a,b]$.

Inner Product for polynomials

Note

- For $p(x)$ and $q(x)$ with at most degree n:

$$\langle p(x), q(x) \rangle = \mathrm{p(0)q(0)} + \mathrm{p(1)q(1)} + \cdots + \mathrm{p(n)q(n)}$$

- For $p(x)$ and $q(x)$: $\langle p(x), q(x) \rangle = \mathrm{p(0)q(0)} + \int_{-1}^{1} p'q'$

- For $p(x)$ and $q(x)$: $\langle p(x), q(x) \rangle = \int_{0}^{\infty} p(x)q(x)e^{-x}dx$

05

Inner Product Space

Inner product space

Definition

An inner product space is a finite-dimensional real or complex vector space V along with an inner product on V.

Euclidean Space Unitary Space

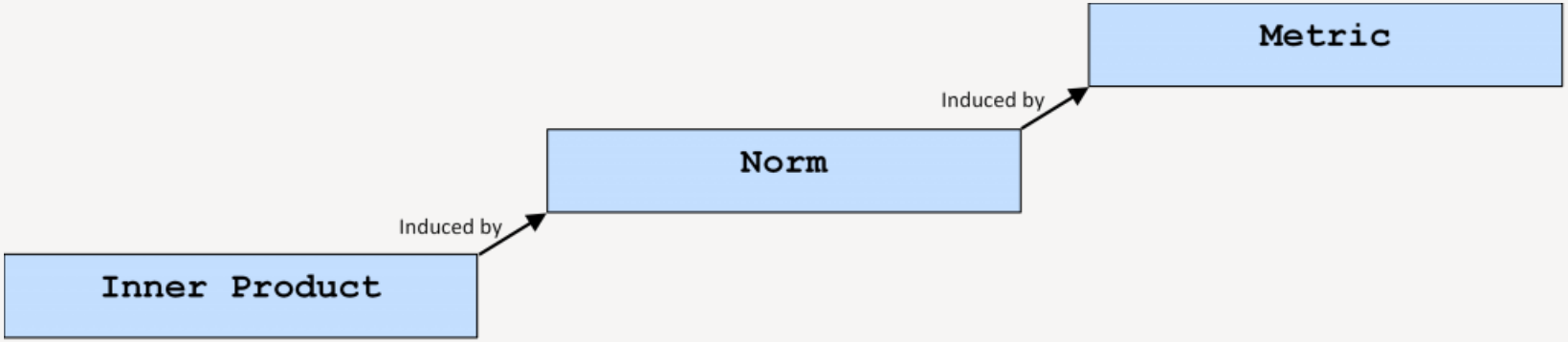
06

Norm Space

The reason to use norms

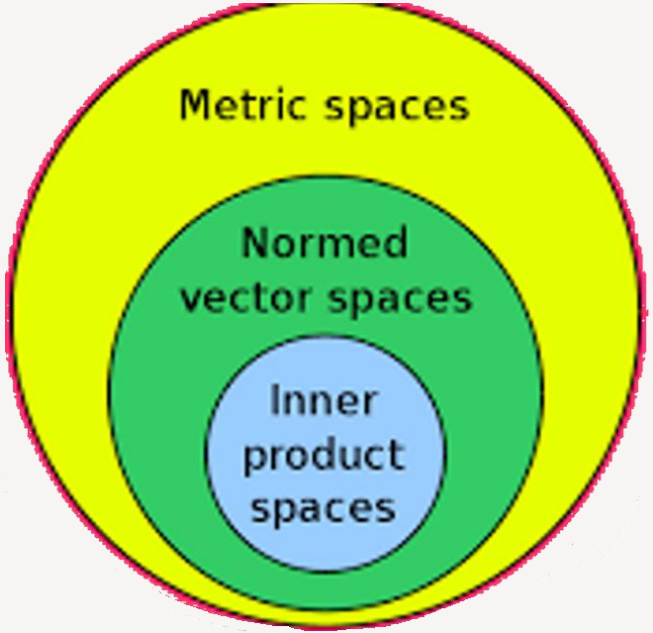
- Machine learning uses vectors, matrices, and tensors as the basic units of representation
- Two reasons to use norms:
  - To estimate how **big** a vector/matrix/tensor is
    - How big is the difference between two tensors is
  - To estimate how **close** one tensor is to another
    - How close is one image to another

Inner Products, Norms and Metrics



Inner Products, Norms and Metrics

- Given an inner product <A , B>, one can obtain a norm doing $|| A ||^2 = <A , A>$
- And given a norm $|| A ||$, one can obtain a metric using the difference vector $|| A - B ||$



Inner Products, Norms and Metrics

Vector space	Generalization
metric	metric space
norm	normed
scalar product	inner product space

Vector Norm Properties

Important Properties

1. Absolute Homogenity / Linearity:

$$||\alpha x|| = |\alpha| \, ||x||$$

2. Subadditivity / Triangle Inequality:

$$||x + y|| \leq ||x|| + ||y||$$

3. Positive definiteness / Point separating:

$$if \; ||x|| = 0 \; then \; x = 0$$
$$(from \; 1 \; \& \; 3): For \; every \; x, ||x|| = 0 \; iff \; x = 0$$

4. Non-negativity:

$$||x|| \geq 0$$

Inner product and norm

Theorem 4

Take any inner product $\langle \cdot, \cdot \rangle$ and define $f(x) = \sqrt{\langle x, x \rangle}$. Then f is a norm.

Note

Every inner product gives rise to a norm, but not every norm comes from an inner product. (Think about norm 2 and norm max)

Proof

- **Positive definiteness**

  Inner product properties give:

  $\langle x, x \rangle \geq 0$ for all x, so $f(x) = \sqrt{\langle x, x \rangle} \geq 0$.

  $\langle x, x \rangle = 0$ iff $x = 0$.

  Hence $f(x) = 0 \iff x = 0$.

- **Homogeneity (absolute scalability)**

  For any scalar α,

  $f(\alpha x) = \sqrt{\langle \alpha x, \alpha x \rangle} = \sqrt{\alpha \, \bar{\alpha} \, \langle x, x \rangle} = \sqrt{|\alpha|^2 \, \langle x, x \rangle} = |\alpha| \sqrt{\langle x, x \rangle} = |\alpha| \, f(x)$.

  (For real spaces, $\alpha \, \bar{\alpha} = \alpha^2$.)

Proof (cont')

- **Triangle inequality**

  Use Cauchy–Schwarz:

  $$|\langle x, y\rangle| \leq f(x)\, f(y).$$

  Now expand:

  $$f(x+y)^2 = \langle x+y, x+y\rangle = \langle x, x\rangle + \langle y, y\rangle + \langle x, y\rangle + \langle y, x\rangle$$

  In real spaces, $\langle y, x\rangle = \langle x, y\rangle$,so

  $$f(x+y)^2 = f(x)^2 + f(y)^2 + 2\langle x, y\rangle \leq f(x)^2 + f(y)^2 + 2|\langle x, y\rangle|$$

  $$\leq f(x)^2 + f(y)^2 + 2f(x)f(y) = \big(f(x) + f(y)\big)^2$$

  Taking square roots (both sides ≥ 0):

  $$f(x+y) \leq f(x) + f(y).$$

- All three norm axioms hold, so f is a norm. □

Euclidean Norm

Definition

Functions closely related to inner products are so-called **norms**. Norms are specific functions that can be interpreted as a distance function between a vector and the origin.

Definition

For $v \in V$, we define the eculidean norm of v, denoted $||v||$, by:

$$||v|| = \sqrt{< v, v >}$$

Euclidean Norm

Note

- Euclidean Norm (2-norm, l_2 norm, length)

$$||x|| = \sqrt{x_1^2 + x_2^2 + \ ... + x_n^2} = \sqrt{x^T x}$$

  - A vector whose length is 1 is called a unit vector
  - **Normalizing**: divide a non-zero vector by its length which is a unit vector in the same direction of original vector
  - It is a nonnegative scalar
  - In $\mathbb{R}^2$ follows from the Pythagorean Theorem.
  - What about $\mathbb{R}^3$?
  - What is the shape of $||x||_2 = 1$?

Euclidean Norm

Example

Norm of $P_n(x)$ in the term of inner product $\langle p_n(x), q_n(x)\rangle = \int_0^1 p_n(x)q_n(x)dx$:

$$||P_n(x)|| = \sqrt{\int_0^1 P_n^2(x)dx}$$

Norm of sum

Theorem 5

If x and y are vectors:

$$||x+y|| = \sqrt{||x||^2 + 2\,x^T y + ||y||^2}$$

Proof:

$$||x+y||^2 = (x+y)^T(x+y)$$

$$= x^T x + x^T y + y^T x + y^T y$$

$$= ||x||^2 + 2\,x^T y + ||y||^2$$

Metric Properties

Important Properties

Let $\mathbf{V}$ be a real vector space over $\mathbb{R}$. A function $\mathbf{V}\times\mathbf{V} \to \mathbb{R}$ is called metric or **distance function** on $\mathbf{V}$, and $(\mathbf{V}, \mathbf{R})$ a metric space, if for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbf{V}$ the following holds true:

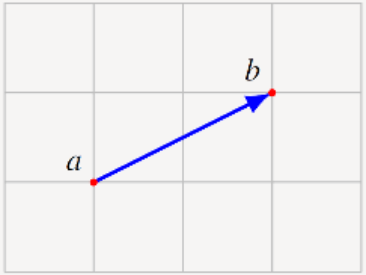
(i) $d(v, w) \geq 0$ and $d(v, w) = 0$ if and only if $v = w$;

(ii) $d(v, w) = d(v, w)$;

(iii) $d(v, w) \leq d(v, u) + d(u, w)$.

Euclidean Distance

- Distance between two n-vectors shows the vectors are "close" or "nearby" or "far".



- Distance:

$$dist(a, b) = ||a - b||$$

Comparing Norm and Distance

Norm

(Normed Linear Space)

1. $\left|\left|x-y\right|\right| \geq 0$
2. $\left|\left|x-y\right|\right| = 0 \Rightarrow x = y$
3. $\left|\left|\lambda(x-y)\right|\right| = |\lambda| \left|\left|x-y\right|\right|$

Distance Function

(Metric Space)

1. $dist(x,y) \geq 0$
2. $dist(x,y) = 0 \Rightarrow x = y$
3. $dist(x,y) = dist(y,x)$

Resources

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- Chapter 2: David Poole, Linear Algebra: A Modern Introduction. Cengage Learning, 2014.
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